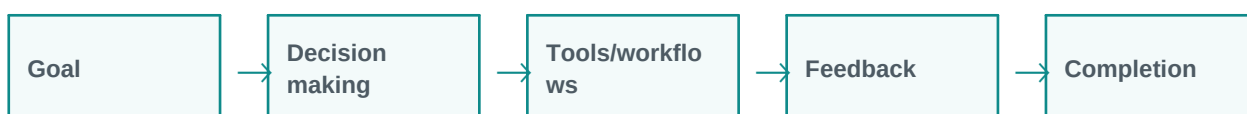


Agentic AI

Agentic AI executes goal-driven workflows.

Simple explanation: Agentic AI is AI that does not only answer questions. It can pursue a goal, make decisions, use tools, follow a workflow, and adapt based on results.

Learning style: this handout starts with a very simple explanation and gradually increases the depth. It is designed for classroom training, participant reading, and quick revision.



What you will learn

- Meaning in simple words
- Key components and architecture
- Practical examples and use cases
- Where it fits in GenAI and agentic AI systems
- Benefits, challenges, and implementation points



1. Beginner-Friendly Explanation

Agentic AI is AI that does not only answer questions. It can pursue a goal, make decisions, use tools, follow a workflow, and adapt based on results.

Think of it like this:

Agentic AI is like a goal-driven assistant. It understands the objective, breaks it into steps, uses tools, checks progress, and moves toward completion.

Simple real-time examples

- Resolve support ticket
- Generate report from files
- Monitor alerts
- Prepare meeting notes
- Handle invoice workflow

2. Gradually Deeper Explanation

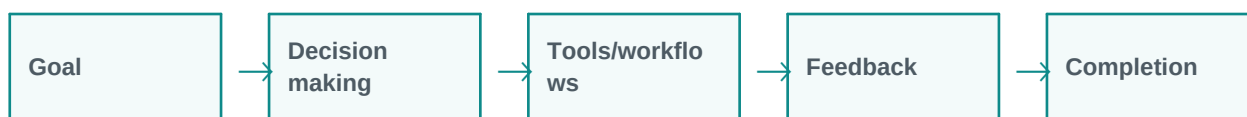
Agentic AI becomes more powerful when we understand its components, data flow, limitations, and business fit. In real projects, the concept is rarely used alone. It usually works with prompts, models, tools, documents, APIs, evaluation, monitoring, and human review.

Important concepts

Concept	Meaning
AI copilots	AI copilots is an important part of understanding and applying Agentic AI in practical GenAI systems.
Automation agents	Automation agents is an important part of understanding and applying Agentic AI in practical GenAI systems.
Operations assistants	Operations assistants is an important part of understanding and applying Agentic AI in practical GenAI systems.
Decision support agents	Decision support agents is an important part of understanding and applying Agentic AI in practical GenAI systems.
Workflow orchestrators	Workflow orchestrators is an important part of understanding and applying Agentic AI in practical GenAI systems.

3. Architecture / Flow

A typical architecture can be understood as a simple flow. The exact design changes based on the project, but the pattern below is a useful starting point.



Architecture explanation

Step 1: Goal - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 2: Decision making - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 3: Tools/workflows - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 4: Feedback - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 5: Completion - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Common implementation layers

- User interface or API layer
- Prompt and instruction layer
- Model layer
- Tool/retrieval/data layer
- Validation, safety, and logging layer
- Human review or feedback layer

4. Detailed Examples

Below are example scenarios showing how this topic appears in real GenAI projects.

Scenario	How it works	Expected output
Resolve support ticket	The system receives the request, applies Agentic AI, and processes the task step by step.	A useful, structured, reviewable result.
Generate report from files	The system receives the request, applies Agentic AI, and processes the task step by step.	A useful, structured, reviewable result.
Monitor alerts	The system receives the request, applies Agentic AI, and processes the task step by step.	A useful, structured, reviewable result.
Prepare meeting notes	The system receives the request, applies Agentic AI, and processes the task step by step.	A useful, structured, reviewable result.



Scenario	How it works	Expected output
Handle invoice workflow	The system receives the request, applies Agentic AI, and processes the task step by step.	A useful, structured, reviewable result.

5. Benefits and Challenges

Benefits	Challenges / Risks
Improves productivity and reduces repetitive manual effort.	Can produce incorrect or unsupported answers if not validated.
Helps users interact with complex systems in natural language.	Needs careful design of prompts, data access, permissions, and monitoring.
Can support training, support, coding, analysis, and document workflows.	May have cost, latency, privacy, and governance concerns in production.

6. Best Practices

- Start with a simple prototype before building a complex system.
- Use clear prompts and structured output formats.
- Add validation and human review for important decisions.
- Measure quality with realistic test cases, not only demo examples.
- Monitor cost, latency, accuracy, safety, and user feedback.
- Document assumptions, limitations, and escalation paths.

7. Key Takeaways

Agentic AI is an important concept in modern LLM and GenAI systems. The simple idea is easy to understand, but production implementation requires thoughtful architecture, data quality, evaluation, safety, and monitoring.

Quick revision

- Simple meaning: Agentic AI is AI that does not only answer questions. It can pursue a goal, make decisions, use tools, follow a workflow, and adapt based on results.
- Architecture: input, processing, tools/data, validation, output.
- Business value: saves time, improves knowledge access, and supports automation.
- Risk control: use grounding, evaluation, guardrails, monitoring, and human approval.